In this document, there are 2298 words outside of math formulas.

Contents

A. Separating the Dynamics from the initialisation of a stochastic process

Fix a domain for stochastic processes $D \subseteq \mathbb{R}^N$ and a time index set $I \subset [0, \infty)$ (so I can be discrete as well, $I = \mathbb{N}$).

On a measurable space $(\Omega, \mathcal{F})$ a **stochastic process** is an I-indexed sequence of random variables with values in D, i.e. a map $X : I \times \Omega \to D$ where $X(t, \cdot)$ is measurable at each fixed time $t \in I$.

This setup already fixes a starting position — the random variable X_0 with its law $\mathbb{P}_{X_0}$. So there are two aspects to this definition that we need to separate conceptually.

Recall that a Markov chain is a stochastic process X satisfying the simple Markov property

$$\forall t_1 < \ldots < t_{k+1} \in I : \quad \mathbb{P}_{X_{t_{k+1}} \mid X_{t_1}, \ldots, X_{t_k}} = \mathbb{P}_{X_{t_{k+1}} \mid X_{t_k}}.$$

A Markov process induces a kernel (*transition matrix* if D is discrete)

$$\text{for any } t \in I, \quad \begin{aligned} &P_t : D \times \mathcal{F} \to [0, 1] \\ &P_t(x, A) := P(X_t \in A \mid X_0 = x), \end{aligned}$$

but a kernel on its own does not determine a stochastic process, not at least until a *starting point* (or distribution) is specified — different processes started at different points can share the same kernel (dynamics). In discrete time and space, it is known that a transition matrix with a choice of a starting point induces a Markov chain.

A stochastic differential equation of the form $dX_t = b(t, X)dt + \sigma(t, X)dw_t, X_0 = x_0$ may be solved by a stochastic process. While the equation + the starting point are what determine the solution, and we are used to saying that the expression $dX_t = b(t, X)dt + \sigma(t, X)dW_t$ is meaningless on its own, it is useful to treat the two as conceptually standalone pieces of the definition — akin to how the Markov kernel is a standalone object paired with each Markov chain. After all, the same b, σ induce different solutions for different starting points x_0.

B. Markov chain action on the space of distributions

C. The sampling problem

As described in [Andrea], we consider 3 classes of sampling problems depending on what is known:

	Task: **produce a sample** $x \sim \mu$ where	
	we know	but do not know
Case I.	the distribution μ	
Case II.	a family of distributions and a sample sequence $(\mu_\theta)_\theta$ and $(x_i)_i \overset{\perp}{\sim} \mu_\theta$	the parameter θ, and hence which μ_θ
Case III.	a sample sequence only $(x_i)_i \overset{\perp}{\sim} \mu$	the distribution μ

We will work in case I. for now and assume we know the distribution μ. Diffusion based sampling is the approach of approximating μ by a sequence $(\nu_t)_t$

D. A generalized time derivative

We want to construct a differential equation describing the evolution of the laws of Y_t. Inspired from PDEs on regular functions of the form $\partial_t p(t,x) = F[t, x, \partial_x, \ldots]$ in mind, we might be tempted to simply replace the function p by the law $\mathbb{P}|Y_t\rangle$ and develop an equation for $\partial_t \mathbb{P}|Y_t\rangle$ (if exists). This naïve way leads to inconsistencies that we outline below.

In particular, a well-behaved generalization of $\partial_t p(t,x)$ should be stable under changes to the initial conditions (the law of Y_0). We now show that $\partial_t \mathbb{P}|Y_t\rangle$ is not.

D.1. The problem with $\partial_t \mathbb{P}|Y_t\rangle$

> **Context.** For two random variables ξ, η and a real (rate) $r > 0$ construct the processes
>
> $$X_t = \xi + r\,t \quad \text{and} \quad Y_t = \eta + r\,t, \tag{1}$$
>
> satisfying the SDEs
>
> $$\begin{aligned} dX_t &= r\,dt \\ X_0 &= \xi \end{aligned} \quad \text{and} \quad \begin{aligned} dY_t &= r\,dt \\ Y_0 &= \eta \end{aligned}$$
>
> sharing the same dynamics $dX_t = r\,dt = dY_t$ but having different initial conditions ξ, η.

So we'd want our generalization of $\partial_t p$ to be the same on both processes.

Derivation. Evaluate $\partial_t \mathbb{P}|X_t\rangle|_{t=0}$ on $[a, \infty)$ for $a \in \mathbb{R}$:

$$
\begin{aligned}
\partial_t \, \mathbb{P}(X_t \geq a)|_{t=0} &= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon}(\mathbb{P}(X_\varepsilon \geq a) - \mathbb{P}(X_0 \geq a)) \\
&\overset{(1)}{=} \lim_{\varepsilon \to 0} \frac{1}{\varepsilon}(\mathbb{P}(\xi + r\varepsilon \geq a) - \mathbb{P}(\xi \geq a))
\end{aligned}
$$

(express[1] $(\xi + r\varepsilon \geq a) = (\xi \geq a) \cup (a - r\varepsilon \leq \xi < a)$)

$$
= \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \mathbb{P}(a - r\varepsilon \leq \xi < a), \tag{2}
$$

And similarly for Y:

$$
\partial_t \, \mathbb{P}(Y_t \in [a, \infty))|_{t=0} = \lim_{\varepsilon \to 0} \frac{1}{\varepsilon} \mathbb{P}(a - r\varepsilon \leq \eta < a).
$$

It is evident now that the derivative depends directly on the laws of ξ and η. If the two were chosen according to, for example, $\xi \sim \mathrm{Unif}(0, 1)$ and $\eta \sim \mathrm{Unif}(0.5, 1)$, and we pick $a = 1$, the limits above would evaluate to

$$
\begin{aligned}
&\partial_t \mathbb{P}(X_t \geq 1) = r, \\
&\partial_t \mathbb{P}(Y_t \geq 1) = 2r,
\end{aligned} \tag{3}
$$

because in that case

$$
\begin{aligned}
&\mathbb{P}(1 - r\varepsilon \leq \xi < 1) = r\varepsilon \\
\text{but} \quad &\mathbb{P}(1 - r\varepsilon \leq \eta < 1) = 2r\varepsilon.
\end{aligned}
$$

□

Conclusion. the time derivative of the laws of X_t, Y_t is not invariant to change of initial conditions.

Why does $\partial_t \mathbb{P}|Z_t\rangle$ vary?

It turns out we can easily visualize this phenomenon if we consider

1. the speed at which (the laws of) X_t and Y_t move through $\mathbb{R}$, and,
2. the speed at which individual sets $A \subseteq \mathbb{R}$ gain or lose weight[2] as t varies, under $\mathbb{P}|X_t\rangle$ and $\mathbb{P}|Y_t\rangle$,

---

[1]Having the usual notation for events in mind, $(\xi \geq a) \equiv \{\omega \in \Omega \mid \xi(\omega) \geq a\}$

[2]measure

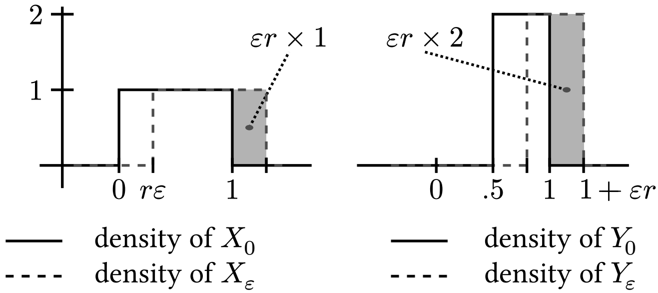


Figure 1: The densities of X and Y at times $t = 0$ and $t = \varepsilon$. Here again $\xi \sim \text{Unif}(0, 1)$ and $\eta \sim \text{Unif}(0.5, 1)$.

and realize that $\partial_t \mathbb{P}|X_t\rangle$ measures the latter, while the invariant quantity is the former.

In Figure 1, the highlighted areas indicate the weight gained by $[1, \infty)$ as time progresses from 0 on. While both X_t and Y_t, as well as their plotted densities, are moving to the right at the same rate r (both driven by a shared $dX = r\,dt = dY$), the rate of weight gain per unit time of $[1, \infty)$ is twice as large in the Y case as in the X case.

D.2. The time-derivative of $\mathbb{P}|X_t\rangle$

> **Context.** Fix a process $Z_t = \zeta + r\,t$ where ζ is a random variable with CDF F_ζ and continuous density $p_\zeta = F_\zeta'$.

Derivation. Pick $a \in \mathbb{R}$ and evaluate the probability gain over $(-\infty, a)$ at t:

$$
\begin{aligned}
\mathbb{P}(Z_t \leq a) - \mathbb{P}(Z_0 \leq a) &= \mathbb{P}(\zeta + r\,t \leq a) - \mathbb{P}(\zeta \leq a) \\
&= \mathbb{P}(\zeta \leq a - r\,t) - \mathbb{P}(\zeta \leq a) \\
&= -\mathbb{P}(a - r\,t \leq \zeta \leq a) \\
&= -\int_{a-r\,t}^{a} p_\zeta(a) da
\end{aligned}
\tag{4}
$$

(by the mean value theorem, for some $a' \in (a - r\,t, a)$)

$$
= -r\,t \times p_\zeta(a').
$$

This gives us a Taylor expansion arount $t = 0$:

$$
\mathbb{P}(Z_\varepsilon \leq a) = \mathbb{P}(Z_0 \leq a) - r\varepsilon \times p_\zeta(a) + o(\varepsilon^2). \tag{5}
$$

The term $r\varepsilon \times p_\zeta(a)$ corresponds to the highlighted rectangles in Figure 1, and where the invariant quantity r is "corrupted" by the multiplier $p_\zeta(a) = F'_\zeta(a)$.

Recall that we can recover the law of a real random variable from its CDF $F(a) = \mu(-\infty, a]$, by the Lebesgue-Stieltjes integral $\mu(A) = \int_A dF(a)$. Denote[3] the measure recovered this way by $\mathbb{P}|F\rangle$. Over all $a \in \mathbb{R}$, Equation 5 is an equation of CDFs, so it naturally induces an equation of *laws*. The (signed) measure induced by the first order term $rp_\zeta(a) = rF'_\zeta(a)$ is then $r\mathbb{P}\big|F'_\zeta\big\rangle$, so Equation 5 in measure speak reads

$$
\mathbb{P}|Z_\varepsilon\rangle = \mathbb{P}|Z_0\rangle - \varepsilon r\,\mathbb{P}\big|F'_\zeta\big\rangle + o(\varepsilon^2).
$$

To recover r when taking the derivative $\partial_t \mathbb{P}|Z_t\rangle$ now it is natural to also divide by $\mathbb{P}\big|F'_\zeta\big\rangle$ in Radon-Nikodym sense:

$$
r = \frac{\mathrm{d}\partial_t \mathbb{P}|Z_t\rangle}{\mathrm{d}\mathbb{P}\big|F'_\zeta\big\rangle},
$$

or, writing it in higher-order derivative notation,

[3]No ambiguity can arise with the notation $\mathbb{P}|\zeta\rangle$ and $\mathbb{P}|F\rangle$ because ζ is of shape $\Omega \to \mathbb{R}^N$, while F is of shape $\mathbb{R} \to \mathbb{R}$. And we have $\mathbb{P}|\zeta\rangle = \mathbb{P}\big|F_\zeta\big\rangle$ if ζ is $\mathbb{R}$-valued.

$$r = \left.\frac{\partial\, \mathrm{d}\mathbb{P}|Z_t\rangle}{\partial t\, \mathrm{d}\mathbb{P}\big|F'_\zeta\big\rangle}\right|_{t=0},$$

which indicates how from the increment of $\mathbb{P}|Z_t\rangle$ we cancel a time t and a density $\mathbb{P}\big|F'_\zeta\big\rangle$ factor. $\square$

That is, we reach the following

> **Conclusion.** A natural initial-distribution-invariant quantity based on the time derivative of Z_t's law is its Radon-Nikodym derivative against the signed measure induced by (the Lebesgue-Stieltjes integral with respect to) Z_0's density.

Two problems arise in the derivation of the last expression: for it to work,
- $Z_0 = \zeta$ must admit a (continuous) density $p_\zeta = F'_\zeta$;
- even then, the absolute continuity condition $\mathbb{P}\big|F'_\zeta\big\rangle \ll \mathbb{P}|Y_t\rangle$ must hold.

Thus it is important to go back to Equation 5 and proceed in a way that does not require canceling a F'_ζ term.

D.3. Getting ~~out of~~ into the box

In Figure 1 we treated the increment $\mathbb{P}|Z_\varepsilon\rangle - \mathbb{P}|Z_0\rangle$ to the first order in ε as a rectangle. We then obtained its width by dividing its mass by its height: $\left(\varepsilon r \times p_\zeta\right)/p_\zeta$. Now, while ζ's density p_ζ might not exist, its mass always does.

And if we write $p_\zeta = 1 \times p_\zeta$, the quantity is numerically identical but probabilistically understandable as a unit weight of ζ, which we always can compute from $\mathbb{P}|\zeta\rangle$. Can we divide the area $\left(\varepsilon r \times p_\zeta\right)$ by $1 \times p_\zeta$ — a unit mass of ζ then, and recover r?

This would correspond to dividing the mass $\left(\varepsilon r \times p_\zeta\right)$ by the mass $\left(1 \times p_\zeta\right)$, thus arriving at a *dimensionless* quantity εr. But that is nonsensical, because εr has units of *space*:

$$Z_\varepsilon = \zeta + r\varepsilon,$$

$$\left.\begin{aligned}[Z_\varepsilon] &= \text{space}\\ [\varepsilon] &= \text{time}\end{aligned}\right\}\ \text{hence}\ \begin{aligned}[r] &= \text{velocity, and}\\ [\varepsilon r] &= \text{space.}\end{aligned}$$

That is, the "modified" division

$$\varepsilon r = \frac{\varepsilon r \times p_\zeta}{1 \times p_\zeta} \quad \left[= \frac{\text{mass}}{\text{mass}}\right]$$

loses a dimension of space. This suggests to add one back, to the numerator. Conceptually this means that instead of as mass over 1D space $\left(\varepsilon r \times p_\zeta\right)$, we consider it as a box $\left(1 \times \varepsilon r \times p_\zeta\right)$ – a mass over 2D space. Having one extra spatial dimension in the domain of the law amounts to including the $Z_0 = \zeta$ variable and considering the *joint* law

$$\mathbb{P}|Z_\varepsilon, Z_0\rangle : \mathcal{B}(\mathbb{R}^2) \to [0, 1].$$

Let us see now how the time increment of $\mathbb{P}|Z_t, Z_0\rangle$ looks like with the last derivation in mind.

D.3.1. Diffusions

> **Context.** Consider the diffusion process
>
> $$Z_t = \zeta + rt + W_t$$
>
> for a constant $r > 0$ and an initial $Z_0 = \zeta$ independent from the Wiener process $(W_t)_t$.

Derivation. Similarly to the diffusion-free context, for any $a \in \mathbb{R}$ we have

$$\begin{aligned}&\mathbb{P}(Z_t \leq a) - \mathbb{P}(Z_0 \leq a)\\ &\qquad = -\mathbb{P}(a - r\,t - W_t \leq \zeta \leq a).\end{aligned}$$

In Equation 4 we expressed the probability as the length of the interval times the density of ζ at a point in it, but in this case, the interval is random – its length depends on W_t. We can still evaluate the joint law, though:

$$
\begin{aligned}
\mathrm{d}\mathbb{P}|Z_0 = x, Z_\varepsilon = z\rangle &= \mathrm{d}\mathbb{P}|\zeta = x, \zeta + r\,t + W_t = z\rangle \\
&= \mathrm{d}\mathbb{P}|\zeta = x, W_t = z - r\,t - x\rangle \\
&\quad \text{(by independence } \zeta \perp W_t\text{))} \\
&= \mathrm{d}\mathbb{P}|\zeta = x\rangle \cdot \mathrm{d}\mathbb{P}|W_t = z - r\,t - x\rangle
\end{aligned}
$$

That is, the law $\mathbb{P}|\zeta\rangle$ naturally falls of from $\mathbb{P}|Z_0, Z_\varepsilon\rangle$, and we cancel ζ's effect completely via the Radon-Nikodym derivative:

$$
\frac{\mathrm{d}\mathbb{P}|Z_0, Z_\varepsilon = y\rangle}{\mathrm{d}\mathbb{P}|Z_0\rangle}(x) = \mathrm{d}\mathbb{P}|W_t = z - r\,t - x\rangle.
$$

On sets $(A, B) \in \mathcal{B}(\mathbb{R}^2)$, the join law decomposition above acts by

$$
\begin{aligned}
\mathbb{P}|Z_0, Z_\varepsilon\rangle(A, B) &= \iint_{(A,B)} \mathrm{d}\mathbb{P}|Z_0, Z_\varepsilon\rangle \\
&= \int_A \left(\int_B \mathrm{d}\mathbb{P}|W_t = z - r\,t - x\rangle \right) \mathrm{d}\mathbb{P}|\zeta = x\rangle \\
&= \iint \underbrace{1_{A,B}(x, x + r\,t + y)\,\mathrm{d}\mathbb{P}|W_t = y\rangle}_{\text{pure evolution}} \mathrm{d}\mathbb{P}|\zeta = x\rangle,
\end{aligned}
$$

where we see the evolution dynamics integrated along the initial condition ζ. The Radon-Nikodym derivative above then corresponds to disintegrating this expression, i.e. taking the integrand only, which is a map

$$
\begin{aligned}
&\mathbb{R} \times \mathcal{B}(\mathbb{R}) \to \mathbb{R} \\
&\quad (x, B) \mapsto \int_B 1_{A,B}(x, x + r\,t + y)\,\mathrm{d}\mathbb{P}|W_t = y\rangle
\end{aligned}
$$

shaped precisely as a Markov kernel. $\square$

Conclusion. While the law of Z_t contains the pure evolution increments in a non-trivial way, extractable (in the $Z_t = \zeta + r\,t$ case) by a Radon-Nikodym derivative against the initial condition's spatial derivative, the joint law of (Z_0, Z_t) contains the entire Z_0 dependence as a clean factor that is easily cancellable by a Radon-Nikodym derivative against $\mathbb{P}|Z_0\rangle$, resulting in

$$\frac{\mathrm{d}\mathbb{P}|Z_t, Z_0\rangle}{\mathrm{d}\mathbb{P}|Z_0\rangle} = \mathbb{P}|Z_t \mid Z_0\rangle.$$

E. The Fokker-Planck equation

> **Context.** While we define[4] a D-valued stochastic process $(Y_t)_t$ on a probability space $(\Omega, \mathbb{P})$ as a map of shape $Y : I \times \Omega \to D$ measurable in the Ω argument, what is *relevant* are all its laws $\mathbb{P}|Y_t\rangle, t \geq 0$, as well as all its joint laws
>
> $$\text{for all } n \text{ and } t_1, ..., t_n, \quad \mathbb{P}\Big|Y_{t_1}, ..., Y_{t_n}\Big\rangle : \mathcal{B}(D^n) \to [0,1].$$
>
> So from the perspective of time flowing, what is relevant is not so much the sequence $(Y_t)_t$ of the *variables*, but rather the sequence $\mathbb{P}|Y_t\rangle_{t\geq 0}$ of their *laws*.
>
> If Y admits probability densities $p(t,x) = \mathrm{d}\mathbb{P}|Y_t\rangle/dx$, it is natural to consider the sequence of densities $(p(t,\cdot)_t)_t$ and ask if their evolution is driven by a partial differential equation
>
> $$\frac{\partial p(t,x)}{\partial t} = F[t, x, p, \partial_x p, \partial_{xx} p, ...], \text{for some } F,$$
>
> and in particular, **what is that F?** In this section, we answer that question positively for Itô diffusions.

In Section D we explained that the proper object to study the distributional evolution of an Itô stochastic process is not $\mathbb{P}|Y_t\rangle$ but $\mathbb{P}|Y_t \mid Y_0\rangle$, we present one approach to deriving the Fokker-Planck equation.

[4]usually for an interval $I \subseteq \mathbb{R}_{\geq 0}$ and a $\mathcal{B}$orel $D \subseteq \mathbb{R}^N$

We will derive it by Tailor-expansion of the transition densities, following the general approach in [Walter] or the referenced inside [Garcia-Palacios], sec. 4 and 5. However, we completely rework it in the language of measures and Markov kernels so that the probability distributions are not assumed to admit densities (against the Lebesgue measure, as done in these resources). This is why we will arrive at a PDE of distributions (weak form) and not of ordinary functions. Only after will we specialize to density-admitting processes (strong form).

E.1. Weak form

We will derive the Fokker-Planck equations by a Taylor expansion of the evolution of $\mathbb{P}|Y_t\rangle$.

> **Context.** Fix a diffusion
>
> $$\begin{aligned} dY_t &= b(t, Y_t)dt + \sigma(t, Y_t)dW_t \\ Y_0 &= y_0 \end{aligned}$$
>
> with b, σ satisfying the relevant linear growth and local Lipschitz conditions so that a solution exists.

Derivation. We first rewrite

$$\mathbb{P}|Y_{t+\varepsilon} \mid Y_t\rangle \overset{(8)}{=} \mathbb{E}\left[\delta_{Y_{t+\varepsilon}} \;\middle|\; Y_t\right]. \tag{9}$$

Now we Taylor-expand the expectand in weak sense

$$\begin{aligned} \forall y' = y + \Delta : \quad \delta_{y'} &= \delta_{y+\Delta} \\ &= \sum_k \delta_y^{(k)} \frac{\Delta^k}{k!} \\ &= \delta_y + \delta_y' \Delta + \delta_y'' \frac{\Delta^2}{2} + \cdots \end{aligned}$$

which evaluated at $y = Y_t$ and $y' = Y_{t+\varepsilon}$ and substituted into Equation 9 yields[5]

References on this page: `("Walter", "Garcia-Palacios")`

Abstract nonsense. In this block we generalize the relation

$$\mathbb{P}[B] = \mathbb{E}[1_B], \quad \text{for every event } B,$$

from events B to random variables X.

On events regarding random variables, $B = (X \in A)$, the equivalence reads

$$\mathbb{P}(X \in A) = \mathbb{E}[1_{X \in A}]. \tag{6}$$

Below we are interested in using that equality over all $\mathcal{B}$orel A, in which case the left hand side can be seen as the law $\mathbb{P}|X\rangle$ evaluated at A. Let us write the right side's dependence on A as a measure, too.

First, realize that the randomness in the expectand in (6) is only through X, so we should be able to write $1_{X \in A}(\omega)$ as a function of $X(\omega)$. Indeed, $1_{X \in A} = 1_A(X)$.

Now recall that the characteristic function $(1_A(x))$ is the Dirac-delta function with flipped arguments: for all $\mathcal{B}$orel A and all $y \in \mathbb{R}$, $1_A(y) = \delta_y(A)$. Evaluating that at $y := X$, $1_A(X) = \delta_X(A)$, we get

$$1_{X \in A} = 1_A(X) = \delta_X(A) \tag{7}$$

and so

$$\forall\ \mathcal{B}\text{orel } A: \quad \mathbb{P}(X \in A) \overset{(6)}{=} \mathbb{E}[1_{X \in A}] \overset{(7)}{=} \mathbb{E}[\delta_X(A)],$$

which in point-free style reads

$$\mathbb{P}|X\rangle = \mathbb{E}[\delta_X] \tag{8}$$

as an equality of measures.

Abstract nonsense. Since the main operation we want to perform — take time-derivatives — is not generally available for measures, we turn our measures into distributions in Schwarz sense. We indicate the Schwarz distribution corresponding to a probability law by dropping the probability sign, i.e. for each random variable X we define a distribution $|X\rangle$ via $\mathbb{P}|X\rangle$ simply by

$$\forall \varphi \in C_0^\infty: \quad \langle \varphi \mid X \rangle = \int \varphi(x)\, \mathrm{d}\mathbb{P}|X = x\rangle,$$

[5] by $\mathbb{E}\left[\delta_{Y_t}^{(k)} \mid Y_t\right] = \delta_{Y_t}^{(k)}$ for all k.

$$\left|Y_{t+\varepsilon} \mid Y_t\right\rangle = \delta_{Y_t} + \delta'_{Y_t}\mathbb{E}[\Delta Y_t \mid Y_t] \\ + \frac{1}{2}\delta''_{Y_t}\mathbb{E}\left[\Delta Y_t^2 \mid Y_t\right] + \cdots \tag{10}$$

Denote the coefficients behind the derivatives of δ_{Y_t} by

$$a_t^{(k)}(y) = \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\mathbb{E}\left[(Y_{t+\varepsilon} - Y_t)^k \;\middle|\; Y_t = y\right]$$

(as almost-everywhere defined functions on $\mathbb{R}$, for each $k \in \mathbb{N}$ and $t \geq 0$).

Glossing over details, we evaluate the coefficients to be

$$\begin{aligned} a_t^{(1)} &= \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\mathbb{E}[Y_{t+\varepsilon} - Y_t \mid Y_t] \\ &= \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\mathbb{E}\left[\int_t^{t+\varepsilon} b ds + \int_t^{t+\varepsilon} \sigma dW_s \;\middle|\; Y_t\right] \\ &\left(\text{since } \mathbb{E}\left[\int \sigma dW_s\right] = 0 \text{ and } \tfrac{1}{\varepsilon}\int_t^{t+\varepsilon} b ds \to b(t)\right) \\ &= b(Y_t, t) \end{aligned}$$

and

$$\begin{aligned} a_t^{(2)} &= \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\mathbb{E}\left[(Y_{t+\varepsilon} - Y_t)^2 \mid Y_t\right] \\ &\left(\text{since } \left(\int b ds\right)^2 = o(\varepsilon^2) \text{ and } ()\right) \\ &= \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\mathbb{E}\left[\left(\int_t^{t+\varepsilon} \sigma(s, Y_s) dW_s\right)^2 \;\middle|\; Y_t\right] \\ &\qquad (\text{Itô isometry}) \\ &= \lim_{\varepsilon\to 0}\frac{1}{\varepsilon}\mathbb{E}\left[\int_t^{t+\varepsilon} \sigma^2(s, Y_s) ds \;\middle|\; Y_t\right] \\ &= \sigma^2(t, Y_t), \end{aligned}$$

while higher-order coefficients $a_t^{(k)}, k > 2$ consist of limits of expectations of linear combinations of terms

$$\frac{1}{\varepsilon}\left(\int_t^{t+\varepsilon} b(t, Y_t)ds\right)^{k-i}\left(\int_t^{t+\varepsilon} \sigma(t, Y_t)dW_s\right)^i$$
$$= \frac{1}{\varepsilon} o\left(\varepsilon^{k-i}\varepsilon^{i/2}\right)$$

which all vanish.

Then Equation 10 with $\partial_\varepsilon|_{\varepsilon=0}$ applied reads

$$\partial_\varepsilon \big|Y_{t+\varepsilon} \mid Y_t\big\rangle \,|_{\varepsilon=0} = b(t, Y_t)\delta'_{Y_t} + \frac{1}{2}\sigma^2(t, Y_t)\delta''_{Y_t},$$

or, introducing explicit variable for $Y_t = y$,

$$\partial_\varepsilon|t + \varepsilon \mid t : y\rangle \,|_{\varepsilon=0} = b(t, y)\delta'_y + \frac{1}{2}\sigma^2(t, y)\delta''_y,$$

to signify that this is an equality of distributions parametrized by y for almost every $y \in \mathbb{R}$, i.e. an equality of objects of shape $\mathbb{R} \underset{\text{a.e.}}{\to} (C_0^\infty \to \mathbb{R})$.

Now we will integrate the free-hanging y against $\mathbb{P}|Y_t\rangle = \mathbb{P}_Y|t : y\rangle$, which on the left side turns

$$\mathbb{P}_Y|t + \varepsilon|t : y \rangle\!\rangle\, t : y\rangle = \mathbb{P}_Y|t + \varepsilon\rangle$$

and so we're left with an equation of distributions

$$\begin{aligned}\partial_\varepsilon|t + \varepsilon\rangle|_{\varepsilon=0} &= \big|b(t, y)\delta'_y + \tfrac{1}{2}\sigma^2(t, y)\delta''_y \rangle\!\rangle\, t : y\big\rangle. \\ &= \big|b(t, y)\delta'_y \rangle\!\rangle\, t : y\big\rangle \\ &\qquad + \frac{1}{2}\,\big|\sigma^2(t, y)\delta''_y \rangle\!\rangle\, t : y\big\rangle.\end{aligned}$$

As distributions,

$$\big|b(t, y)\delta'_y \rangle\!\rangle\, t : y\big\rangle = -\partial_y\left[b(t, y)|t : y\rangle\right]$$
$$\big|\sigma^2(t, y)\delta''_y \rangle\!\rangle\, t : y\big\rangle = \partial_y^2\left[\sigma^2(t, y)|t : y\rangle\right],$$

which can be easily derived by acting on test functions[6]. The right hand sides here are the distributional derivatives of the products of b and σ^2 with the distribution of Y_t, $|t : y\rangle$.

And so we derived the Fokker-Planck equation:

$$\begin{aligned} \partial_\varepsilon |t+\varepsilon\rangle|_{\varepsilon=0} = -\partial[b(t,y)|t : y\rangle] \\ +\frac{1}{2}\partial^2\left[\sigma^2(t,y)|t : y\rangle\right] \end{aligned} \qquad \square$$

We can spell the result in a less clumsy way if we leave the bound variable y implicit. For that we need to write the drift and the diffusion coefficients with the time argument as an index: $b_t(y), \sigma_t(y)$ instead of $b(t,y), \sigma(t,y)$.

Conclusion. Let $(Y_t)_t$ be the solution of the SDE

$$\begin{aligned} dY_t &= b_t(Y_t)dt + \sigma_t(Y_t)dW_t \\ Y_0 &= y_0 \end{aligned}$$

where $b_t(y), \sigma_t(y)$ satisfy the linear growth and local Lipschitz conditions.

Then its laws $\mathbb{P}|Y_t\rangle$ satisfy the Fokker-Planck equation:

$$\partial_\varepsilon |Y_{t+\varepsilon}\rangle|_{\varepsilon=0} = -\partial[b_t|Y_t\rangle] + \frac{1}{2}\partial^2\left[\sigma_t^2|Y_t\rangle\right]. \tag{11}$$

E.2. Strong form

Now we specialize to the case when $(Y_t)_t$ has all densities.

[6] For any $\varphi \in C_0^\infty$,

$$\begin{aligned} \langle\varphi \mid b(t,y)\delta_y' \rangle\!\rangle\; t : y\rangle &= \langle b(t,y)\varphi \mid \delta_y' \rangle\!\rangle\; t : y\rangle \\ &= -\langle b(t,y)\dot\varphi \mid t : y\rangle \\ &= \langle\dot\varphi \mid -b(t,y) \mid t : y\rangle \\ &= \langle\varphi \mid \partial \mid -b(t,y) \mid t : y\rangle \end{aligned}$$

> **Context.** Let $(Y_t)_t$ be the solution of the SDE as above, and additionally assume that its laws admit densities against the Lebesgue measure, i.e.
>
> $$\forall t \geq 0 : \quad \mathrm{d}\mathbb{P}|Y_t\rangle = p_t(y)dy$$

Susbstitute $|Y_t\rangle = p_t(y)dy$ in Equation 11 to get

$$\partial_\varepsilon p_{t+\varepsilon}(y)dy|_{\varepsilon=0} = -\partial[b_t p_t dy] + \frac{1}{2}\partial^2\left[\sigma_t^2 p_t dy\right].$$

Now simply read out the classical PDE in fron of dy:

Conclusion. The Fokker-Planck equation on the densities p_t of $(Y_t)_t$:

$$\partial_t p_t = -\partial_y(b_t p_t)(y) + \frac{1}{2}\partial_y^2(\sigma_t^2 p_t)(y)$$

E.2.1. Chapman-Kolmogorov

> **Context.** A real-valued Markov process $(Y_t)_t$.

For two time points $t_1 \leq t_2$ and an $\varepsilon > 0$, the Chapman-Kolmogorov equation for the process Y states that the probability flux between times t_1 and $t_2 + \varepsilon$ goes through all possible values at the midtime t_2:

$$\mathbb{P}|Y_{t_2+\varepsilon} \mid Y_{t_1}\rangle = \mathbb{P}|Y_{t_2+\varepsilon} \mid Y_{t_2} \rangle\!\rangle\, Y_{t_2} \mid Y_{t_1}\rangle.$$

Now we are interested to derive the time-evolution of the left-hand side, and so we need only two time points. Fix $t > 0$ and $\varepsilon > 0$ and plug $t_1 = t_2 = t$ above:

$$\mathbb{P}|Y_{t+\varepsilon} \mid Y_t\rangle = \mathbb{P}|Y_{t+\varepsilon} \mid Y_t \rangle\!\rangle\, Y_t \mid Y_t\rangle$$

and integrate along $|Y_t\rangle$

$$\mathbb{P}|Y_{t+\varepsilon}\rangle = \mathbb{P}|Y_{t+\varepsilon} \mid Y_t \rangle\!\rangle\, Y_t\rangle,$$

or, if we allow ourselves to write the time indices only:

Conclusion. For any Markov $(Y_t)_t$ and times $t, t+\varepsilon$,

$$\mathbb{P}_Y|t+\varepsilon\rangle = \mathbb{P}_Y|t+\varepsilon \mid t \rangle\!\rangle\, t\rangle.$$

F. Ornstein-Uhlenbeck

Let $x \sim \mu$ such that $x \perp W$.

$$\begin{aligned} dZ_t &= -Z_t dt + \sqrt{2} dW_t \\ Z_0 &= x \end{aligned}$$

From $e^{-t} d e^t Z_t = dZ + Z dt$ we find the solution is

$$\begin{aligned} Z_t &= e^{-t} x + \sqrt{2} e^{-t} \int_0^t e^s \, \mathrm{d}W_t \\ &= e^{-t} x + \sqrt{1 - e^{-2t}}\, G_t, \quad (x, G_t) \sim \mu \otimes \mathcal{N}(0,1) \end{aligned}$$

where we rescaled the stochastic integral to

$$G_t := \left(1/2(e^{2t} - 1)\right)^{-1/2} \int_0^t e^s \, \mathrm{d}W_s$$

to get a unit normal[7] $G_t \sim \mathcal{N}(0,1)$.

G. The $\nabla \log p$ term

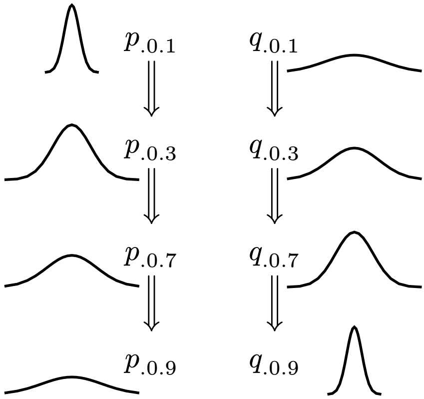
Pick an SDE

$$dX_t = b_t(X_t) dt + \sigma_t dW_t, \quad X_0 \sim \mu$$

and put its distributions $p_t dx = d\mathbb{P}_{X_t}$.

[7] $\int_0^t e^s \, \mathrm{d}W_s$ is normal of variance $V = \int_0^t e^{2s} \, \mathrm{d}s = \frac{1}{2}(e^{2t} - 1)$, thus $G_t := V^{-1/2} \int_0^t e^s \, \mathrm{d}W_s$ is normal of variance 1.

Diffuse	**Denoise**
$dX = bdt + \sigma dW$	$dY = \tilde{b}dr + \tilde{\sigma}dW$



Now let us find $\tilde{b}, \tilde{\sigma}$ that make for a process Y_r having the densities of X_t in reverse order.

So we are after $\tilde{b}, \tilde{\sigma}$ such that $dY_r = \tilde{b}_r(Y_r)dr + \tilde{\sigma}_r dW_r$ has densities $d\mathbb{P}_{Y_r} = d\mathbb{P}_{X_t}$.

Let $q_r dx = d\mathbb{P}_{Y_r}$. Then we desire $q_r = p_t$ for all $r \sim t$.

Both p_t, q_r satisfy the respective Fokker-Plancks

$$\begin{aligned} \dot{p_t} &= F[b_t, \sigma_t, p_t] \qquad &&\text{for all } t \in I \\ \dot{q_r} &= F\left[\tilde{b}_r, \tilde{\sigma}_r, q_r\right] \qquad &&\text{for all } r \in I \end{aligned}$$

To connect the $\tilde{b}, \tilde{\sigma}$ to b, σ we need to couple the two equations. Clearly we can get a relationship between $\dot{p}$ and $\dot{q}$ if we ∂_r the equality $q_r = p_t$ (considering $r = r(t)$, we have $\dot{r}$):

$$\dot{q}_r = \partial_r q_r \overset{!}{=} \partial_r p_t = \dot{p}_t \partial_r t = \dot{t}\dot{p}_r.$$

Naturally, if we pick $r \sim t$ to be $r + t = T$, the relation is $\dot{q}_r = -\dot{p}_t$, meaning that p_t flows oppositely to q_t. Applying Fokker-Planck now gives

$$F\left[\tilde{b}_r, \tilde{\sigma}_r, q_r\right] = \dot{t}F[b_t, \sigma_t, p_t].$$

$$\nabla \cdot \left[\tilde{b}_r q_r\right] - \frac{1}{2}\Delta\left[\tilde{\sigma}_r^2 q_r\right] = \dot{t}\left(\nabla \cdot [b_t p_t] - \frac{1}{2}\Delta\left[\sigma_t^2 p_t\right]\right)$$

We have two terms of shape $\nabla \cdot (p_t \cdot)$ and two of shape $\Delta(p_t \cdot)$; rearrange

$$\nabla \cdot \left[p_t\left(\tilde{b}_r - \dot{t}b_t\right)\right] = \frac{1}{2}\Delta\left[p_t\left(\tilde{\sigma}_r^2 - \dot{t}\sigma_t^2\right)\right]$$

By recalling that $\Delta = \operatorname{div} \circ \operatorname{grad}$, pull a div from both sides:

$$\operatorname{div}\left[p_t\left(\tilde{b}_r - \dot{t}b_t\right)\right] = \operatorname{div}\left[\frac{1}{2}\operatorname{grad}\left[p_t\left(\tilde{\sigma}_r^2 - \dot{t}\sigma_t^2\right)\right]\right].$$

Put $s_t \equiv \tilde{\sigma}_r^2 - \dot{t}\sigma_t^2$ for short. Split cases $p_t = 0$ and $p_t > 0$ in the right hand side to pull out a p_t term

$$\nabla[p_t s_t] = \left(1_{p_t > 0} + 1_{p_t = 0}\right)\nabla[s_t p_t]$$

Assuming we pick $s_t > 0$, note that on $p_t = 0$ we have[8] $\nabla s_t p_t = 0$, so the $1_{p_t=0}$ term vanishes:

$$\begin{aligned} &= p_t \frac{\nabla s_t p_t}{p_t} 1_{p_t > 0} \\ &= p_t s_t 1_{p_t > 0} \nabla[\ln s_t p_t] \end{aligned}$$

The equation has one obvious solution for $\tilde{b}_r$ now if we drop the div and divide by p_t:

$$\tilde{b}_r = \dot{t}b_t + \frac{1}{2}s_t 1_{p_t > 0}\nabla[\ln p_t s_t] \quad \text{for all } r \sim t.$$

For a saner appearance, pick $\tilde{\sigma}_r = \sigma_t$ and $r + t = T$ so that $\dot{t} = -1$ and $s_t = 2\sigma_t^2$, then

[8] p_t is non-negative, so at $x \in D$ with $p_t(x) = 0$, the gradient must be zero, otherwise $p_t(x - \varepsilon \nabla p_t(x)) < 0$ for sufficiently small $\varepsilon > 0$

$$\tilde{b}_r = -b_t + \sigma_t^2 1_{p_t>0} \nabla\left[\ln \sigma_t^2 p_t\right].$$

This derivation works if σ_t is given strictly positive for all times everywhere. If σ_t hits zero, the world shatters.

If $\sigma_t(x) = \sigma_t$ is constant in space, then[9] $\nabla\left[\ln \sigma_t^2 p_t\right] = \nabla[\ln p_t]$ and

$$\tilde{b}_r = -b_t + \sigma_t^2 \nabla[\ln p_t] 1_{p_t>0}.$$

[9] $\nabla\left[\ln \sigma_t^2 p_t\right] = \nabla\left[\ln \sigma_t^2 + \ln p_t\right] = \cancel{\nabla[\ln \sigma_t^2]} + \nabla[\ln p_t] = \nabla[\ln p_t]$